

Student portal: workflow audit

Huchu schools vertical audit · Student portal: workflow audit · student-portal/workflows.md · rendered 2026-09-14

Surface: `students.<tenant> host`, routed to `app/portal/student/**`, backed by `app/api/v2/schools/portal/student/**`, `lib/schools/student-day-loader.ts`, and the shared routes `/api/v2/schools/assignments`, `/api/v2/schools/goals`, `/api/notifications`. Audience: pupils with `SchoolStudent.userId` linked to a `User` of role `STUDENT`.

Audit date: 2026-09-14. Method: static read of the code on `main`, cross-checked against the roadmap in `docs/expansion-plan/schools-roadmap.md`, the prototype contract `docs/design-system/portals/student.html`, and the K-12 benchmark in `../reference/k12-benchmark.md`. Companion UI/UX document: `ui-ux.md` in this folder.

1. Verdict in one paragraph

The student portal does what its twelve `done` stories claim: a pupil sees today's periods, the week's timetable, published marks with movement against last term, homework with a hand-in flow, per-subject goals, a self-service library, and a notification inbox. Identity is scoped to the signed-in pupil with hard 403s. Two things undercut it. First, unpublished marks leak through the subjects and goals routes, which contradicts the whole moderation and publishing chain. Second, the sign-in and account story the prototype promises (student ID plus PIN, forgotten PIN, biometrics, theme, notification cadence) does not exist, and the settings screen links to a password page the proxy bounces back. Homework hand-in is text or a pasted link, not a file. The portal is honest about what is not built, which is to its credit, but five roadmap stories remain open and several prototype screens have no story at all.

Runtime check (14 September 2026)

Verified on the seeded St Marys tenant as `student@stmarys.test` (see `../reference/runtime-verification.md`). B1 confirmed at runtime: after the head rolled term marks into a DRAFT sheet, `/me/subjects` returned a `currentMark` for 13 of 13 subjects while `/me/results` returned zero published lines. B2 confirmed: the password link lands back on the portal home. The library, homework and goals screens rendered as described.

2. Docs versus code

Claim	Source	Code reality	Verdict
S-6.22 timetable today and week, S-6.23 open a class: <code>done</code>	roadmap	<code>student-timetable-screen.tsx</code> , <code>me/timetable</code> .	True.
S-6.24 assignment list with filters, S-6.25 hand in with handed-in state: <code>done</code>	roadmap	<code>student-homework-screen.tsx:156-191</code> posts <code>{action: "submit"}</code> with text and <code>attachmentUrl</code> . No file upload. LATE is recorded, not refused.	True as coded; the prototype's "attach" is a URL field.

Claim	Source	Code reality	Verdict
S-6.26 marks for the term: done	roadmap	me/results filters PUBLISH ED .	True.
S-6.28 to S-6.30 library browse, borrow, return, scan, renew, reserve: done	roadmap	me/library GET/POST. A pupil can create a loan from the phone with issuedById set to the pupil, with no librarian step.	True; integrity question noted in §5.
S-6.32 goals: done	roadmap	student-goals-screen.tsx over /api/v2/schools/goals. goalsForStudent reads result lines with no sheet-status filter.	True, with a leak.
S-6.34 profile edit and theme: done	roadmap	Profile is read-only with "who owns each field"; settings says theme is "not here yet" (student-settings-screen.tsx:79-84).	Story marked done, not built.
S-6.35 help with helpfulness feedback: done	roadmap	Static FAQ. No feedback control found.	Partly built.
S-6.36 own mobile shell: done	roadmap	student-portal-shell.tsx, four tabs.	True.
S-6.20/21 PIN sign-in and recovery, S-6.27 report card, S-6.31 fines payment, S-6.33 cadence: todo	roadmap	Absent. Login is email + password.	Accurate.
Prototype: Messages row on profile, first-run tour, signed-out screen, biometric option	student.html	Absent; no story.	Contract gap not tracked.
Profile copy: Settings has "Alerts, theme and your PIN"	student-profile-screen.tsx:37	None of the three exist.	Copy contradicts behaviour.
Help copy: only published marks are shown	student-help-screen.tsx:23-29	True for the Marks screen, false for Goals.	Copy contradicts behaviour on one screen.

3. Workflow inventory

#	Workflow	Status	Evidence and gap
S1	Get an account (invite, claim, set password)	Implemented	Same invite engine as parents; token handed over by hand.
S2	Sign in (email + password)	Implemented	Prototype wants student ID + PIN.
S3	Forgot password / change password	Broken	Settings links /settings/profile; proxy.ts:469-479 redirects any STUDENT request outside /portal/student back to the portal. No reset route for pupils.
S4	Home: today's periods, next class	Implemented	student-day-loader.ts.

#	Workflow	Status	Evidence and gap
S5	Timetable day and week; open a class	Implemented	
S6	Homework list with filters	Implemented	Published assignments for the pupil's class or stream.
S7	Hand in homework	Partial	Text and URL only; can resubmit until marked; no file upload; no teacher feedback thread (API supports <code>feedback</code> and <code>sendBack</code> , UI shows score only).
S8	Marks for the term with delta	Implemented	PUBLISHED only.
S9	Report card download	Missing	S-6.27 <code>todo</code> ; document source exists (<code>schools.report-card</code>) but STUDENT cannot call the render route.
S10	Goals per subject	Implemented	Leaks unpublished current mark (§5).
S11	Library: browse, borrow, return, scan, renew, reserve	Implemented	Self-issue without a librarian.
S12	Pay a library fine	Missing	S-6.31 depends on S-7.3.
S13	Notifications inbox, mark read, archive	Implemented	Generic <code>/api/notifications</code> . Nothing school-side emits to a pupil except notices.
S14	Notification cadence and quiet hours	Missing	S-6.33 depends on S-9.5.
S15	Profile view	Implemented	Read-only by design.
S16	Edit profile, theme	Missing	S-6.34 wrongly <code>done</code> .
S17	Help centre	Implemented	Static. No "did this help" control.
S18	Messages with teachers	Missing	Prototype profile row; no story.
S19	Learning resources / materials	Missing	<code>teaching-resources GET</code> needs <code>schools.academics view</code> ; STUDENT lacks it.
S20	Exam timetable	Missing	Only <code>SchoolCalendarEvent Kind.EXAM</code> .
S21	Attendance view (own)	Missing	Data exists per student; no student route.
S22	Fees view (own balance)	Missing	Legacy aggregate returns invoices; new shell has no screen. Product decision: many schools do not show fees to pupils.
S23	Boarding leave request	Missing	<code>SchoolLeaveRequest create</code> requires <code>boarding:approve-leave</code> ; boarders cannot request from the portal.
S24	Report a concern / wellbeing	Missing	No model.

#	Workflow	Status	Evidence and gap
S25	Clubs, sport, houses	Missing	No model.
S26	Sign out	Implemented	

Counts: 11 implemented, 1 partial, 1 broken, 13 missing.

4. Benchmark gap

Graded against `../reference/k12-benchmark.md` §17, §19, §13.

Capability	Benchmark norm	Huchu today	Priority
Assignment submission with file upload and teacher feedback	Standard	Text/link, score only	P1
Learning materials per subject	Standard	Absent for pupils	P1
Own attendance summary	Standard	Absent	P2
Report card in app	Standard	Absent (S-6.27)	P1
Exam timetable and seat	Common	Absent	P2
Messages to teacher	Common	Absent	P2
Simple sign-in for younger pupils (ID + PIN)	Common	Absent	P1
Boarding leave request by the boarder	Common in boarding schools	Absent	P2
Wellbeing / report a concern	Increasingly standard	Absent	P2
Library self-service	Common	Present, arguably too permissive	Done
Goals and progress	Uncommon; a differentiator	Present	Done

5. Bugs and risks

#	Finding	Location	Severity
B1	Security and correctness: unpublished marks visible to pupils. <code>currentMark</code> in <code>me/subjects</code> reads result lines for the term with no <code>sheet.s</code> <code>tatus = PUBLISHED</code> filter; <code>goalsForStudent</code> does the same. A DRAFT or HOD_REJECTED mark shows on Goals.	<code>app/api/v2/schools/port al/student/me/subjects/route.ts:68-71</code> ; <code>lib/schools/goals-meetings.ts:110-114</code>	Medium
B2	Dead "Change your password" link.	<code>components/schools/port al/student/student-settings-screen.tsx:44</code> ; <code>proxy.ts:469-479</code>	Medium
B3	Copy promises PIN, alerts and theme that do not exist.	<code>student-profile-screen.tsx:37</code>	Low

#	Finding	Location	Severity
B4	Library self-issue: a pupil creates a loan against a copy with no librarian confirmation, moving stock in the system.	me/library/route.ts:210-233	Low
B5	Legacy aggregate GET /api/v2/schools/portal/student returns guardians' phone and email to the pupil and is unused by the shell. Delete.	app/api/v2/portal/_handlers.ts:365-560	Low
B6	No audit trail for hand-ins, loans or goal edits.	lib/schools/audit.ts	Low
B7	Roadmap row S-6.34 marked done for work not present.	docs/expansion-plan/schools-roadmap.md	Doc

6. Proposed edits

1. **Fix B1 now.** Add `sheet: { status: "PUBLISHED" }` to both queries and a test in `goals-meetings.test.ts` that a DRAFT line does not surface. This is a one-line change that restores the publishing guarantee.
2. **Give pupils a password path.** Either build a portal-local "change password" under `/portal/student/settings` or remove the link and add the "ask the office" copy the teacher portal already uses.
3. **Correct copy** on the profile screen (B3) and the roadmap row S-6.34.
4. **Homework feedback.** The API already carries `feedback` and `sendBack`; show them on the hand-in card so the pupil sees why work came back.
5. **Library issue confirmation.** Make a pupil's borrow a `RESERVED_FOR_PICKUP` state that the librarian confirms at the desk, or keep self-issue but mark it visibly as "collect from the library" until scanned.
6. **Delete the legacy aggregate route** (B5).

7. Proposed restructuring

- **Sign-in.** Replace email + password with student number + PIN for the student host (S-6.20/21). Store the PIN as a bcrypt hash on the linked `user`; add a "forgotten PIN" that the class teacher or office resets from the student record. This is the biggest usability win for younger pupils and shared devices.
- **Home should lead with what is due.** The prototype's home has "Your next class", "This week" KPIs (homework to hand in, latest mark), and recent marks. The code's home is today's periods only. Add the homework-due count and the latest published mark to Home.
- **Merge Notifications and school news.** Notices arrive through the generic inbox; the prototype shows a "School news" banner on Home. One inbox with a Home teaser is enough.
- **Settings is the right place for account, alerts, appearance, privacy** (prototype's five groups). Keep the honest "not here yet" rows until each lands, but wire the two that are trivially available: browser push preference (already exists for teachers) and reduced motion.

8. Proposed new workflows and features

1. **Report card in the portal (S-6.27).** Reuse `schools.report-card` with a student-scoped render branch (same fix as the parent download, see parent audit B1).
2. **File upload on hand-in.** `lib/uploads/` and the record files engine exist (S-4.2). Add `SchoolAssignmentSubmission.fileId` and accept an upload from the phone camera.
3. **Learning materials.** A student-scoped read of `SchoolTeachingResource` filtered to the pupil's class subjects and `sharedWithPupils = true`.
4. **Own attendance** over `students/[id]/attendance` with the portal resolver.
5. **Messages to teachers** once S-7.1 lands; pupil to subject teacher only, visible to the office.
6. **Boarding leave request by the boarder**, parent-approved, warden-actioned; reuses `SchoolLeaveRequest` with a new `REQUESTED_BY_STUDENT` origin.
7. **Exam timetable** as a first-class `SchoolExamSession` with room and seat, surfaced in the timetable tab during exam weeks.
8. **Report a concern** (wellbeing) to a named staff role with restricted visibility. Sits with the parked behaviour/discipline decision.
9. **Fine payment** rides on S-7.3.

9. Open decisions

- Should pupils ever see fee balances? The legacy aggregate exposes invoices; the new shell does not. Decide and delete the other path.
- Self-issue library loans: allowed, or reserve-then-confirm?
- Is the student host meant for primary pupils (PIN, no email) as well as secondary?